

Financial pressure requires a monthly response

Financial Health Analysis / BI10 Round 01 / Task 2

−13.35

points in mean health
909 matched November–December
customers

45

customers ever stressed and very highly
engaged
Health <40 and engagement ≥80

681

December customers in the priority support
cohort
74.18% of 918 observed customers

What the evidence shows

Annual mean health is **66.37**, but **84.10%** of December customers score below 60. The deterioration also appears within the same customers.

Transaction frequency rises **95.82%** in the matched sample. Average ticket size rises only **0.11%**.

Business recommendation

Offer voluntary, tailored support to **25 stressed customers** and scalable budgeting tools to **656 monitoring customers spending above income**.

Eligibility follows observed financial pressure. Engagement informs service design and requires separate contactability checks.

Decision proposed: validate a consent-based support pilot. Intervention impact and ROI remain unmeasured.

Four required analyses, supported by an audit trail

01	Case context and analytical scope	03
02	Financial health distribution and monthly deterioration	04–05
03	Factors associated with low health	06–07
04	Differences by age, province and occupation	08–10
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Scope: the complete Financial Health Analysis deliverable for Task 2. This module supports the wider team submission for Tasks 1–5.

Monthly observations provide the right unit of analysis

Business question

ITB seeks to understand customer financial wellbeing and identify opportunities for supportive services.

This analysis covers health-score distribution, associated factors, demographic differences and a reproducible health–engagement crossover rule.

Source and coverage

Dataset	Coverage
Consumer-month	10,992 records / 999 customers
Raw transactions	1,852,394 records
Calendar	12 months in 2025
Geography	34 provinces and cities

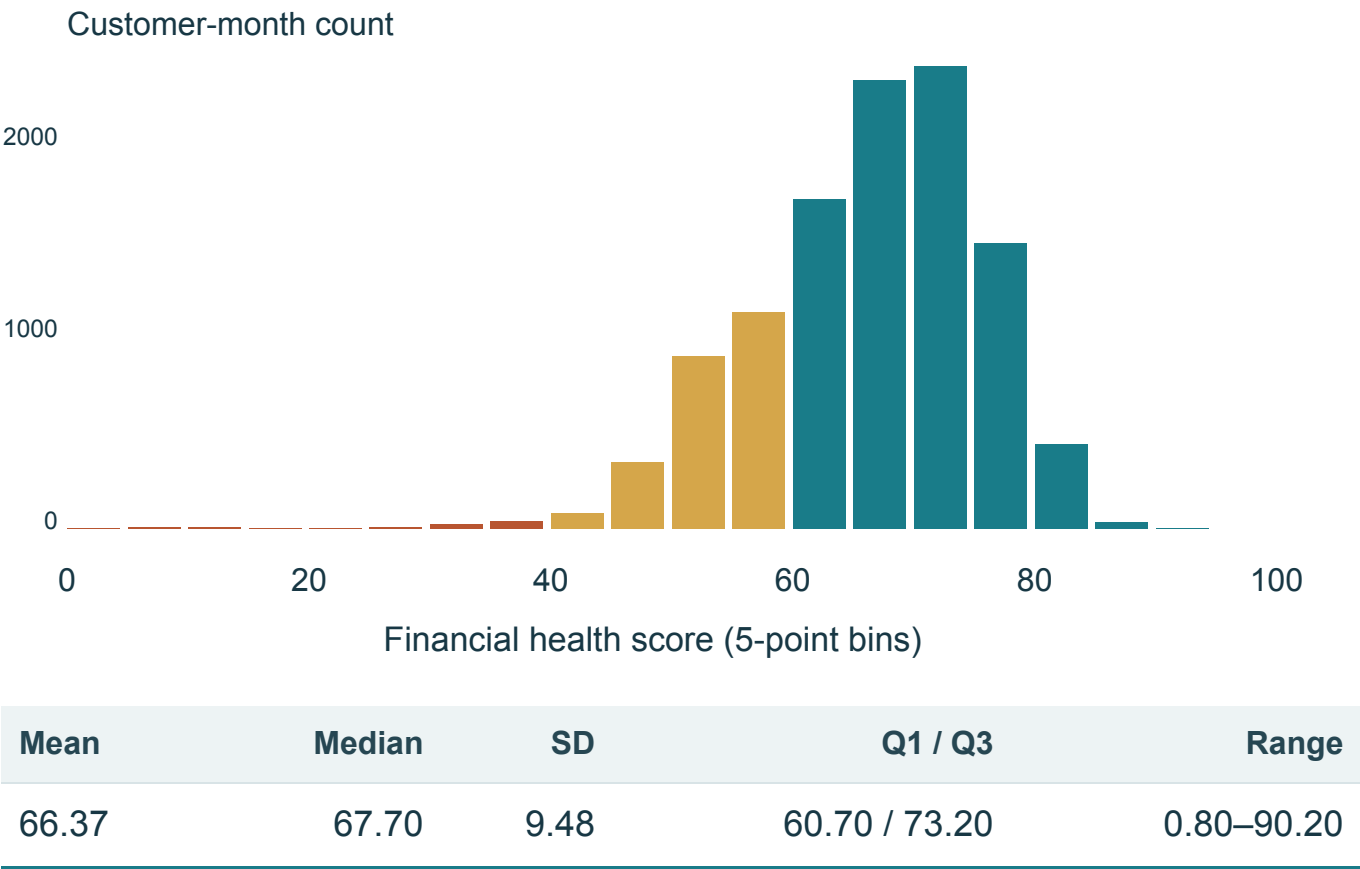
Denominators answer different questions

Question	Denominator
Annual prevalence	10,992 customer-months
Ever-qualified reach	999 distinct customers
Current support reach	918 December customers
Paired change	909 matched customers
Balanced-panel check	908 customers with 12 months

Coverage: 908 customers have 12 months, 86 have one month and five have two months. An unobserved month does not imply zero spend.

Use ratios for comparisons across customers. Synthetic income, credit limits and balances do not represent realistic VND magnitudes.

23.22% of observed months fall below a health score of 60



Health bands

Band	Months	Share
Stressed <40	95	0.86%
Monitoring 40–<60	2,457	22.35%
Stable 60–<80	7,965	72.46%
Healthy ≥80	475	4.32%

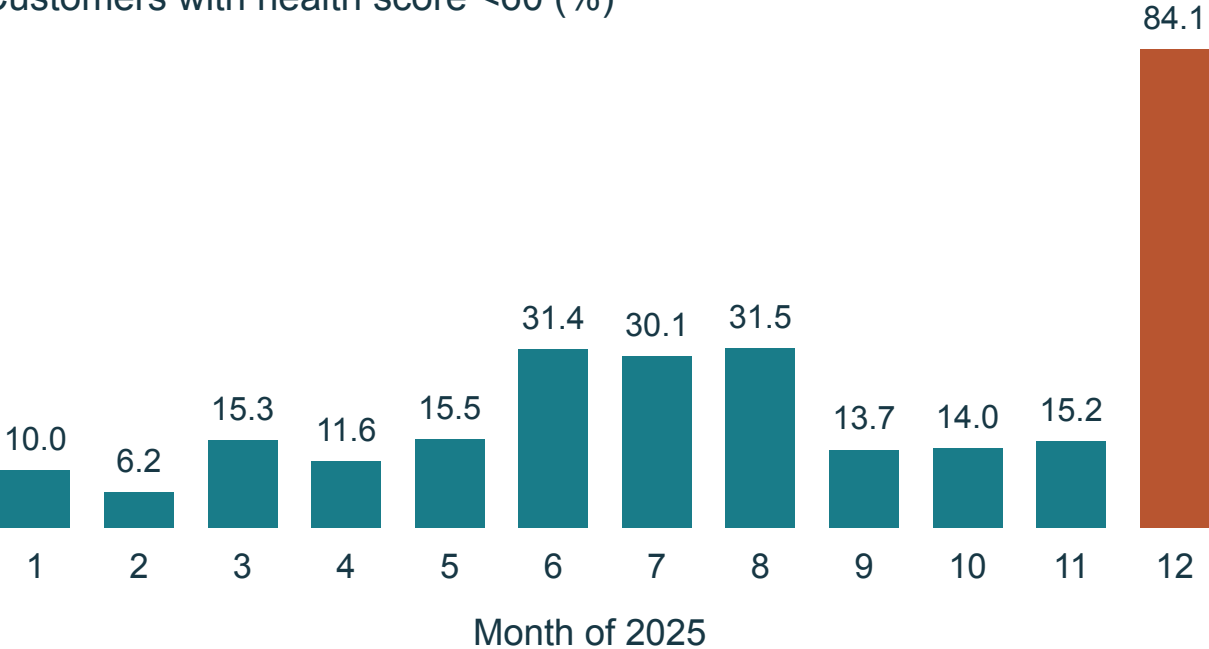
70 of 999 customers (7.01%) experience at least one stressed month.

No customer has an observed-year average below 40. Annual averaging therefore misses all 70 customers with episodic stress.

The stable band dominates annual observations, while customer-level stress can be temporary.

December deterioration persists in a matched-customer comparison

Customers with health score <60 (%)



December snapshot: **772 / 918 customers (84.10%)** score below 60. Monthly samples range from 911 to 920 customers.

-13.35

mean health points

Matched mean: **67.52 → 54.16**

Bootstrap 95% interval: **-13.92 to -12.74**

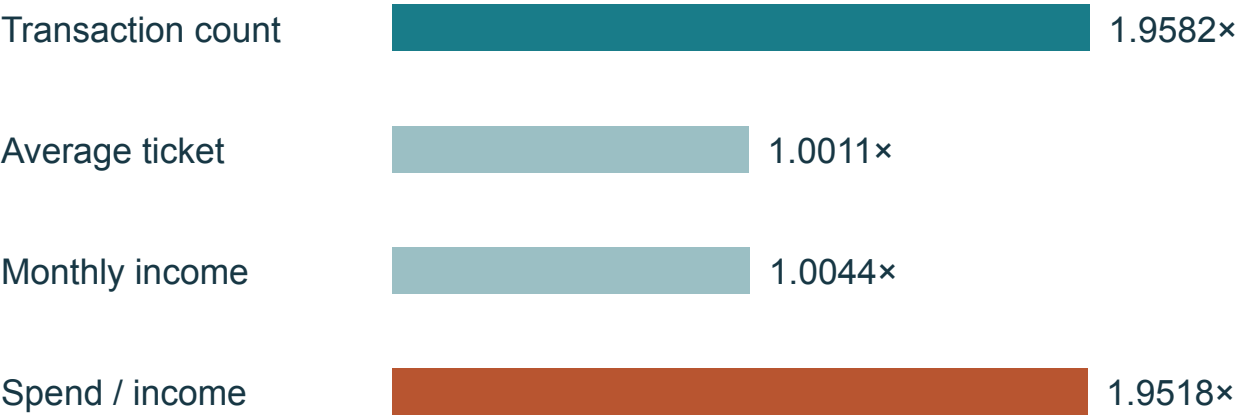
909 matched customers	Result
Health declines	840 (92.41%)
Below 60 in November	139 (15.29%)
Below 60 in December	769 (84.60%)
Stable/healthy → below 60	642 / 770 (83.38%)

The complete 908-customer panel shows a similar **-13.41-point** change.

One synthetic year supports an observed December pattern. It cannot establish recurring seasonality.

Higher transaction frequency accounts for the pressure increase

December / November geometric multipliers



Category evidence

All **14 categories** contribute positively to the mean customer spending increase.

In-store groceries contributes **15.83 percentage points**, or **15.00%** of total mean growth.

The largest absolute category-share shift is only **0.53 percentage points**.

The evidence supports more frequent spending across categories. It does not isolate a shift toward discretionary purchases.

Spend / income = count × exact ticket / income

Frequency +95.82%, ticket +0.11%, income +0.44%. The decomposition is an accounting identity, not a causal explanation of intent.

Spend relative to income has the strongest pooled association

Correlation with financial health score

Indicator	Pearson	Spearman	Excl. December	Within customer & month
Spend / income	-0.898	-0.947	-0.902	-0.850
Credit utilization	-0.862	-0.874	-0.851	-0.842
Spending volatility	-0.496	-0.525	-0.518	-0.659
Essential spend share	+0.481	+0.474	+0.527	+0.783

Interpretation

Spend/income and utilization show the strongest pooled associations among **13 screened indicators**. Their directions persist across robustness checks.

Essential share has a much weaker correlation across customer averages: **+0.079**, versus **+0.481** across months.

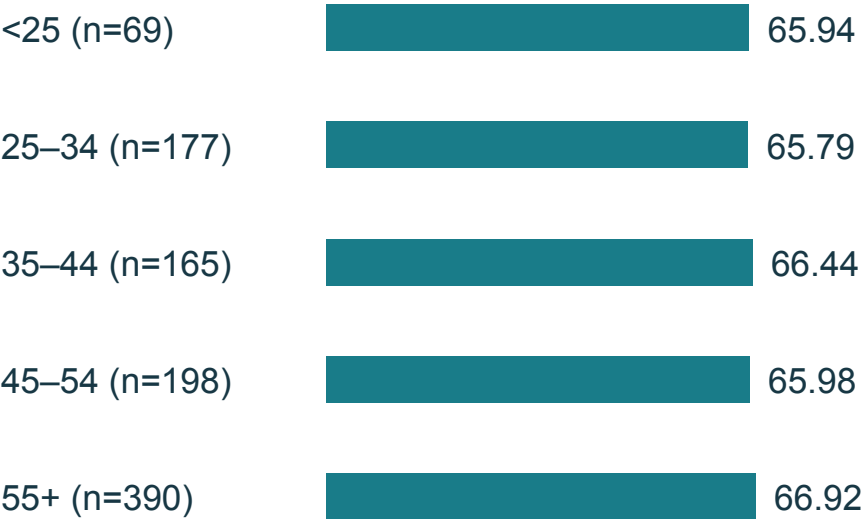
Limits on attribution

Spend/income and utilization share a spending numerator. Conditional VIFs are approximately **10.8 and 10.7**.

The score formula is unavailable. Associations may partly reflect score construction and do not establish independent causal effects.

The primary analysis keeps all observations. Trimming the outer 1% of each predictor separately produces no sign reversals.

Age differences are small relative to the monthly deterioration



Mean score on a 0–100 scale. n = distinct customers.

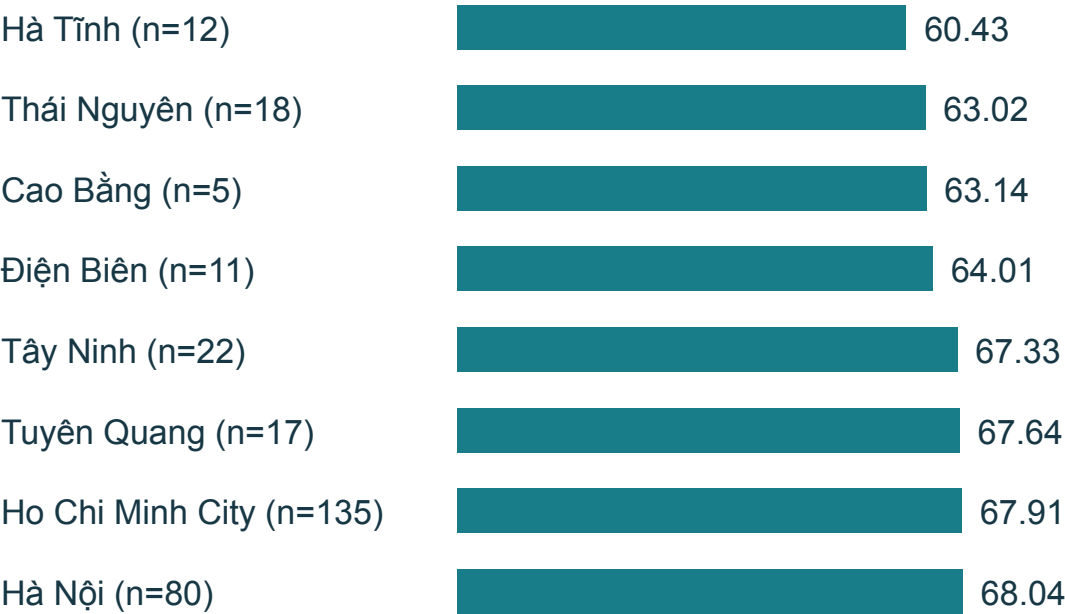
Age	Pooled mean	Equal-customer mean	Below 60
<25	65.94	65.75	23.14%
25–34	65.79	65.79	24.48%
35–44	66.44	66.44	21.87%
45–54	65.98	65.98	24.27%
55+	66.92	66.74	22.64%

The pooled gap between the lowest and highest age groups is **1.13 points**.

Under equal-customer weighting, the lowest group changes from **25–34** to **under 25**.

Age offers descriptive context. The evidence does not justify age-based support exclusions.

Provincial gaps warrant investigation with sample size in view



7.61

points between the extremes

Ha Tinh: 60.43, 12 customers

Hanoi: 68.04, 80 customers

The same extremes remain under equal-customer weighting.

These are unadjusted comparisons. Customer mix, incomplete coverage and small provincial samples limit generalization.

Four lowest and four highest pooled means. Scale: 0–100.

Province identifies areas for further diagnosis. Eligibility should continue to follow observed financial pressure.

Occupation rankings depend on very small groups

Occupation	Customers	Months	Mean health
Film-set designer	1	12	49.80
Barista	1	12	50.83
Environmental education officer	1	12	52.28
Legal secretary	1	1	54.50
Lawyer	3	14	76.35
Chief technology officer	2	24	77.20
Chief operating officer	3	36	77.44
Marketing director	3	25	77.72

Four lowest and four highest pooled means. English translations of source occupation labels.

396

occupation labels across 999 customers

128 occupations have one customer.

No occupation has more than **10 customers**.

The 49.80 to 77.72 range describes this sample. It cannot establish occupational population differences.

Occupation is unsuitable as an eligibility rule here. Full results are retained in the embedded demographic evidence.

45 customers are stressed and very highly engaged in at least one month

financial_health_score < 40 AND engagement_score ≥ 80

Apply both conditions to the same customer-month. Health <40 is the stressed band. Engagement ≥80 is the supplied very-high-engagement band.

52 / 10,992 qualifying customer-months 0.4731% of annual observations	45 / 999 distinct ever-qualified customers 4.5045% of annual customers	8 / 918 December customers 0.8715% of current customers
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Observed segment profile

Metric	Mean	Median
Health score	30.14	34.65
Engagement score	88.17	87.80
Spend / income	1.89	1.84

Support implications

All **52 qualifying observations** have spending above income.
Six of 45 customers qualify in two or more months. Recheck eligibility each month instead of assigning a permanent label.

A stricter engagement threshold changes reach substantially

Health <40, varying engagement cutoff

Engagement	Months	Ever customers	December
≥60	93	69	25
≥70	81	64	25
≥80	52	45	8

≥80 gives an interpretable strict segment. The ≥60 and ≥70 alternatives remain visible for sensitivity.

Very high engagement occurs in 54.74% of stressed observations versus 34.79% overall. At ≥70, that comparison reverses (85.26% versus 91.44%).

Eight qualifying December customers

Customer ID	Health	Engagement	Spend / income
KH00110485	9.2	81.5	2.30
KH00109405	14.1	84.9	2.62
KH00113965	19.7	83.2	1.65
KH00102730	24.4	81.4	1.97
KH00105650	35.0	85.5	1.75
KH00110170	36.3	84.6	1.95
KH00117965	37.5	88.6	2.53
KH00111850	37.6	80.4	1.65

Engagement describes transaction activity. It does not establish consent, contactability or willingness to accept support.

681 December customers qualify for priority support

Priority group and rule	Customers	December share	Proposed support
1. Stressed: health <40	25	2.72%	Voluntary budget review and tailored reminders
2. Monitoring: 40≤health<60, spend/income>1	656	71.46%	Opt-in budgeting and cumulative-spend alerts
3. Monitoring: 40≤health<60, spend/income≤1	91	9.91%	Planning tools and financial education
4. Stable / healthy: health≥60	146	15.90%	General self-service tools

Priority cohort

Groups 1 and 2 contain **681 customers (74.18%)**, including **306 with engagement ≥80**.

The four groups are disjoint and cover all **918 December customers**.

Delivery constraints

POS represents **62.96%** of spend for group 1 and **59.24%** for group 2, using equal-customer means.

Confirm usable channels and consent before contact. An app-only workflow has insufficient evidence.

These are observed eligibility counts. They do not represent predicted uptake, proven impact or confirmed delivery capacity.

A randomized pilot can test whether support reduces spending pressure

Proposed design

At the next observable cycle, identify eligible customers using information already available. Confirm consent and contactability.

Randomize customers within each support tier between a support invitation and existing service. Keep standard support available to everyone.

Primary outcome

Next-month share spending above income, analyzed by original randomized assignment.

Secondary outcomes and guardrails

Track spend/income, health-score change, tool activation and opt-out or complaint rates. Report follow-up coverage and missing outcomes by arm.

Conditions before launch

Set the minimum worthwhile effect, power calculation, contact capacity and stopping rules before randomization.

Monthly files support action after month close. In-month alerts need a separately validated live feed.

Health scores must never determine credit approval, credit-limit reductions or account blocking.

Definitions and limitations preserve the meaning of the findings

Analytical definitions

Pooled mean: average across observed customer-months. **Equal-customer mean:** average of each customer's observed-period mean.

Exact ticket: total spend / transaction count. **Geometric multiplier:** $\exp(\text{mean}(\log(\text{December value} / \text{November value})))$.

Utilization: spend / credit limit, capped at 1.5.

Within association: remove customer and month means, add the overall mean in the balanced panel.

Resampling: 3,000 customer bootstrap draws for paired changes, seed 20260928. Demographic and category intervals use 1,000 customer resamples.

Data handling

Retain source observations, including extremes. No imputation or winsorization in the primary analysis. Exclude future labels, IDs, health bands and raw VND magnitudes from factor screening.

Material limitations

Synthetic source dates fold into one 2025 calendar. Density artifacts may contribute to monthly patterns. Findings describe this simulated sample.

The score-generation formula and volatility aggregation window are not fully documented. Demographic comparisons are unadjusted and exploratory.

Source register

- [1] BI10_ROUND01.pdf, PDF pp. 3–4, 9, 18–19.
- [2] consumer_financial_health_case_study.md.
- [3] data_dictionary.xlsx.
- [4] consumer_financial_health_engagement_2025.csv.
- [5] Supplied raw transaction data.

The PDF embeds full demographic tables, crossover records, sensitivity results and analysis scripts as supplementary files. Core evidence remains visible in the report.

Complete province comparison

Province / city	Customers	Pooled	Equal-customer
Hà Tĩnh	12	60.43	59.04
Thái Nguyên	18	63.02	62.98
Cao Bằng	5	63.14	63.14
Điện Biên	11	64.01	61.92
Đà Nẵng	31	64.20	64.66
Gia Lai	15	64.50	64.17
Quảng Trị	12	64.69	64.69
Cà Mau	21	64.79	64.74
Quảng Ngãi	17	64.91	65.27
Lâm Đồng	40	64.91	65.63
Lai Châu	5	65.32	65.32
Lạng Sơn	11	65.48	64.93
Lào Cai	24	65.62	66.05
Thanh Hóa	36	65.94	66.01
Vĩnh Long	29	65.99	66.36
Ninh Bình	28	66.09	66.41
Quảng Ninh	13	66.13	66.13

Province / city	Customers	Pooled	Equal-customer
Đồng Nai	49	66.15	65.93
An Giang	34	66.23	66.50
Khánh Hòa	20	66.32	65.79
Phú Thọ	37	66.36	66.44
Hưng Yên	37	66.41	66.64
Sơn La	19	66.48	66.72
Bắc Ninh	30	66.49	66.49
Nghệ An	36	66.49	66.43
Đồng Tháp	36	66.61	66.19
Đắk Lắk	24	66.72	67.09
Cần Thơ	45	66.89	67.06
Huế	15	66.95	66.06
Hải Phòng	35	66.98	67.00
Tây Ninh	22	67.33	67.33
Tuyên Quang	17	67.64	67.65
Thành phố Hồ Chí Minh	135	67.91	67.59
Hà Nội	80	68.04	67.82

Sorted by pooled mean health score. Both means use the 0–100 scale. Customer counts refer to distinct customers. Unequal month coverage explains differences between weighting methods.

Qualifying customer-month register (1 of 2)

Rule: health <40 and engagement ≥80 in the same month. S / I = spend-to-income ratio.

Customer ID	Month	Health	Eng.	S / I
KH00100765	2025-01	38.2	87.1	1.050
KH00101965	2025-04	35.6	96.6	1.509
KH00102010	2025-02	25.8	88.2	1.770
KH00102730	2025-12	24.4	81.4	1.967
KH00103570	2025-04	34.6	90.7	1.286
KH00103770	2025-02	35.6	90.1	1.829
KH00103770	2025-08	36.9	86.9	1.309
KH00104050	2025-01	32.4	85.5	1.194
KH00104085	2025-01	30.8	83.1	1.464
KH00104130	2025-07	34.5	90.4	2.622
KH00105650	2025-12	35.0	85.5	1.751
KH00105845	2025-01	20.7	87.6	1.566
KH00106010	2025-01	37.8	93.3	1.247

Customer ID	Month	Health	Eng.	S / I
KH00106165	2025-02	32.2	80.5	1.860
KH00106810	2025-02	30.0	88.0	1.221
KH00106810	2025-05	7.9	92.6	2.245
KH00106850	2025-11	34.8	94.5	2.532
KH00107050	2025-05	33.3	95.7	2.417
KH00107565	2025-06	12.6	93.9	2.298
KH00107610	2025-05	38.5	88.1	1.946
KH00107610	2025-07	38.7	83.3	1.490
KH00108210	2025-09	37.0	90.1	2.279
KH00108570	2025-01	34.4	82.9	1.203
KH00108810	2025-01	14.4	87.5	1.632
KH00109405	2025-12	14.1	84.9	2.623
KH00109605	2025-07	38.0	95.7	2.333

52 records across 45 distinct customers. Repeated IDs represent repeated qualifying months. Customer IDs come from synthetic data.

Qualifying customer-month register (2 of 2)

Rule: health <40 and engagement ≥80 in the same month. S / I = spend-to-income ratio.

Customer ID	Month	Health	Eng.	S / I
KH00110170	2025-12	36.3	84.6	1.950
KH00110485	2025-03	6.3	83.2	2.480
KH00110485	2025-12	9.2	81.5	2.296
KH00110970	2025-08	36.8	86.1	2.869
KH00111285	2025-01	35.3	81.4	1.251
KH00111850	2025-06	36.2	94.3	2.092
KH00111850	2025-09	38.7	89.7	1.316
KH00111850	2025-12	37.6	80.4	1.651
KH00111965	2025-08	38.8	87.1	2.705
KH00112130	2025-03	29.7	85.7	2.100
KH00112205	2025-08	35.8	86.2	1.465
KH00113045	2025-01	35.4	92.3	1.311
KH00113965	2025-10	8.2	94.4	3.357

Customer ID	Month	Health	Eng.	S / I
KH00113965	2025-12	19.7	83.2	1.651
KH00114925	2025-10	37.5	95.3	2.492
KH00115130	2025-01	14.8	92.3	2.411
KH00115445	2025-01	35.1	86.0	1.377
KH00115845	2025-01	38.9	94.2	1.165
KH00117525	2025-01	34.7	90.7	1.127
KH00117530	2025-04	33.5	80.6	2.154
KH00117845	2025-09	11.2	89.6	2.505
KH00117965	2025-12	37.5	88.6	2.526
KH00117970	2025-02	38.9	89.9	1.376
KH00118485	2025-05	31.1	83.3	2.165
KH00119010	2025-01	29.5	85.2	1.352
KH00119250	2025-06	32.3	94.9	2.498

52 records across 45 distinct customers. Repeated IDs represent repeated qualifying months. Customer IDs come from synthetic data.